

RESEARCH ARTICLE

Design of a Self-Built Bidirectional Multi-Slope Integrating Analog-to-Digital Converter

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ABSTRACT In order to meet the requirements of high-precision and high-resolution measurement environments, this paper proposes a bidirectional multi-slope integrating ADC based on the Multi-Slope III technique. The paper details the design methodology and operational principles centered on the ‘multiple integration and high-bit sampling integration with low-bit residual detection’ mode, and rigorously derives the theoretical integration waveform along with the dependence of resolution on signal integration time settings. When the signal integration time is set to one power-line cycle (PLC) with a 10 V full-scale DC input, the integrated voltage waveform exhibits stable settling with negligible distortion, confirming effective suppression of nonlinear errors. Combined with the designed optimization schemes such as the “balanced circuit” and “power frequency cycle-matched integration time”, zero-adjustment errors and power frequency interference are effectively suppressed, further improving the measurement resolution to 7.5 bits. The bidirectional integration mechanism adopted expands the input range to -10 V to $+10$ V, eliminating the need for excessive reliance on external signal conditioning circuits. This architecture demonstrates significant utility in low-frequency, high-resolution measurement systems.

INDEX TERMS Analog-digital conversion, high-precision, multi slope, 7.5-digit resolution.

I. INTRODUCTION

In modern electronic systems, analog-to-digital converters (ADCs) serve as critical components. With ever-increasing demands for information processing, ADCs are now required to deliver enhanced precision and resolution, particularly in fields such as medical devices and precision measurement. For instance, strong power frequency interference in power monitoring systems can degrade the quality of original signals and cause distortion of raw data, posing a primary challenge to signal conversion [1]. In the field of medical diagnosis, the fidelity of image grayscale details depends on the precision of ADC [2]. Signals in these domains are typically weak, requiring high-precision and high-resolution ADCs to ensure accurate data acquisition and processing [3], [4].

In recent years, several innovative achievements have emerged in the field of high-precision ADCs. Reference [5]

combines the incremental ADC (IADC) with the extended counting technique of a 9-bit SAR ADC, achieving an effective number of bits (ENOB) of 20.58 in a $0.25\text{-}\mu\text{m}$ CMOS process. Reference [6] proposes a high-resolution programmable-gain incremental $\Delta\text{-}\Sigma$ ADC by integrating triple-sampling technology with an auto-zeroing integrator. Additionally, the multi-slope integrating ADC presented in Reference [7] achieves a balance between conversion rate and power consumption through a three-step conversion cycle and a single active integrator design. Despite it shows unique advantages in the balance of performance and practicability, there is still have significant room for optimization compared with various high-resolution architecture ADCs developed in recent years.

However, for traditional dual-slope/multi-slope ADCs adopting the “single-integration + single-range quantization” scheme, their resolution is constrained by the trade-off between integration time and noise, accompanied by the issue of linearity discontinuity. Specifically, high-bit quantization

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TABLE 1. Comparison of technical limitations and key innovations.

Limitations of Existing Technologies	Novel Solutions Proposed in This Work
The "single integration + single-range quantization" scheme limits resolution	The "multiple integrations + high-bit sampling integration, low-bit residual detection" scheme
Utilize external circuits to suppress noise interference	Integration time is matched to the power frequency cycle
Integral zeroing suffers from overcharging, leading to continuous accumulation of periodic errors	A balancing circuit is designed, with the balanced potential as the "zero" point
Integration time and resolution are mutually restrictive	Employ adjustable integration time to enable dynamic resolution selection

and residual measurement share the same integrating capacitor, causing the nonlinearity from high-bit integration to be directly propagated to residual detection, which makes full-scale linearity errors difficult to overcome. In contrast to current Sigma-Delta ADCs that rely on the synergistic design of oversampling, noise shaping, and digital filtering, and SAR ADCs that utilize external filtering circuits to suppress power-line interference, multi-slope integrating ADCs offer a distinct advantage [8]. Thanks to the inherent “noise averaging” capability of the integration process—the longer the integration time, the more times noise is averaged—multi-slope integrating ADCs can leverage this averaging effect by matching the integration time to the power-line cycle duration, thereby improving the power line frequency rejection ratio (PLFRR) of over 100 dB [9].

This paper presents a bidirectional multi-slope integrating analog-to-digital converter (ADC) based on the Multi-Slope III architecture [10], [11]. Through multiple integrations and a hierarchical design featuring “high-order sampling integration with low-order residual detection”, the proposed ADC achieves both resolution enhancement and improved integrator linearity. A balanced circuit is employed to replace the conventional nulling circuit, effectively preventing the accumulation of zeroing errors caused by overcharging in subsequent measurements. Additionally, a dedicated calibration procedure is initiated at the start of each integration cycle to adjust the reference source and zero potential, thereby eliminating errors induced by common-mode interference and temperature drift [12]. These innovations collectively enable high-precision and high-resolution analog-to-digital conversion. The corresponding technical limitations and key innovations are shown in Table I below.

II. SYSTEM WORKING PRINCIPLE

The bidirectional multi-slope integrating ADC in this system primarily consists of an input buffer, input control switch, reference current source, integrating quantization module, and master controller [13]. The reference current source comprises a high-stability precision power supply, analog

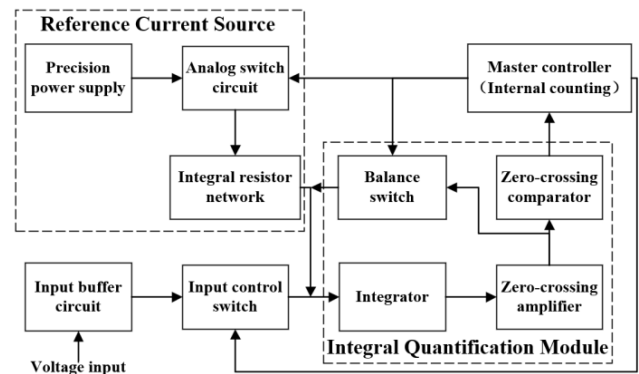


FIGURE 1. Block diagram of bidirectional multi-slope integrating ADC system.

switch circuit and temperature drift compensation integrating resistor network [14], [15], [16], [17], [18]. The integrating quantization module is the core of the analog-to-digital converter, comprising a low-input bias current low-noise integrator, balanced switch, zero-crossing amplifier, and zero-crossing comparator [19], [20]. These modules perform the analog-to-digital conversion of the input signal under the control of the main controller and implement counting within the main controller. The ADC system block diagram is shown in Figure 1.

During ADC operation, the master controller executes phase-synchronized switching sequences to generate precise reference integration currents by regulating the analog switch circuit. Concurrently, the input signal is converted into an input current after impedance isolation via the input buffer circuit, with the master controller governing the input control switch to determine the input signal's integration time. When the input current and reference current are superimposed and fed into the low-input-bias, low-noise integrator, the integrator's output voltage increases or decreases over time. The zero-crossing amplifier then amplifies signals near the zero-crossing point of the integrator output to enhance resolution, ensuring sufficient sensitivity for zero-crossing detection by the comparator. Upon completion of integration, the master controller initiates the auto-balancing phase: activating the nulling path while disconnecting the input switch. This routes the amplifier output through a precision transimpedance network to generate a counteracting current opposing the reference current, forcing the integrator output to settle at a predetermined offset voltage (typically 20.5 mV) above ground potential.

III. CORE CIRCUIT DESIGN

The circuit design of bidirectional multi-slope integrating ADC is mainly divided into reference current source circuit, input control circuit and integral quantization module. The reference current source circuit determines the size of the reference integral current of the input integrator; the input control circuit controls the integration time of the corresponding input signals; the integrating quantization

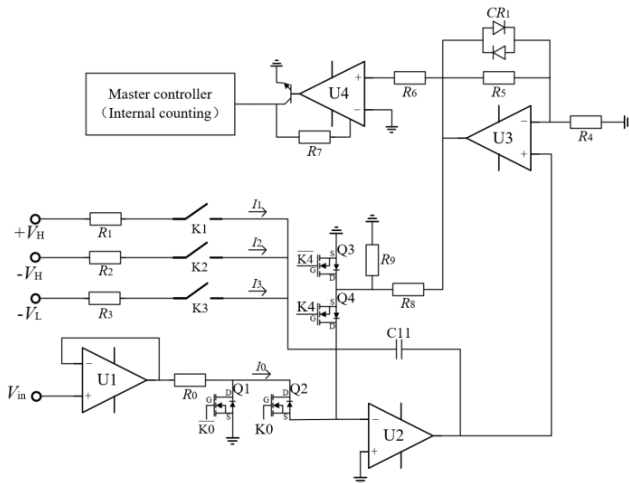


FIGURE 2. Bidirectional multi-slope integrating ADC circuit schematic.

module is the core of the multi-slope integrating ADC, in which the linearity of the integrator, noise rejection, input bias current and other performance indicators are closely related to the accuracy of the ADC. Bidirectional multi-slope integrating ADC circuit schematic is shown in Figure 2 below.

A. REFERENCE CURRENT SOURCE CIRCUIT

The reference current source circuit mainly consists of a precision power supply, an analog switch circuit, and an integral resistance network [21], [22], [23]. $\pm V_H$ and $-V_L$ are high-stability reference voltages generated by the precision power supply, R_1 - R_3 form a low-temperature-drift, high-stability resistor network encapsulated within a single chip. Specifically, $+V_H$ and R_1 , along with $-V_H$ and R_2 , generate the inverse coarse reference current I_1 and forward coarse reference current I_2 of approximately $525 \mu A$ respectively, $-V_L$ and R_3 generate the forward fine reference current I_3 with about $4.1 \mu A$, and the master controller controls the analog switch K_1 - K_3 to realize different reference currents into the integrator. Notably, there is a 128-fold multiplier relationship between the coarse reference current and the fine reference current [24], [25], [26].

B. INPUT CONTROL CIRCUIT

The input control circuit comprises an input buffer U_1 , input control switch Q_1 , Q_2 , among other components. The input buffer isolates the impedance between the input signal and subsequent circuits, converting it into an input current via resistor R_0 . Input control switches regulate the duration during which the input current enters the integrator. The signal integration time T offers selectable values of 0.01 Power Line Cycle (PLC), 0.1 PLC, 1 PLC, or 10 PLC. At an operating frequency of 50 Hz, these correspond to $200 \mu s$, 2 ms, 20 ms, and 200 ms, respectively. The chosen integration time T directly determines the resolution of this bidirectional multi-slope integrating ADC.

C. INTEGRATING QUANTIZATION CIRCUIT

The integrating quantization circuit includes an integrator, a zero-crossing amplifier, a zero-crossing comparator, and a balancing switch. The integrator employs a low-noise precision operational amplifier (ISL28110) with an input bias current as low as 2 pA. The zero-crossing amplifier, composed of a fully differential amplifier U_3 (THP210) with 2-pA input bias current, fast-recovery diode FRD_1 , and associated peripheral circuitry, enhances resolution around the zero-crossing point of the integrated signal [27]. Comparator U_4 compares the amplified signal against ground, generating a high/low logic level that the master controller utilizes for timing and counting operations. The low-power, high-speed balancing switch, formed by N-channel enhancement-mode MOSFETs Q_3 and Q_4 , is driven by complementary control signals. During ADC operation, Q_3 conducts while Q_4 remains off; upon completion, Q_3 turns off and Q_4 activates, biasing the circuit to the balance potential when inactive [28].

IV. FUNDAMENTAL OPERATING PRINCIPLES

In Figure 2, the magnitudes of the coarse reference currents I_1 and I_2 must exceed that of input current I_0 to ensure the full-range input signals into the ADC integrator exhibit monotonic rising or falling transitions. The designed range of signal V_{in} spans from $-10 V$ to $+10 V$, with R_0 configured as $25 k\Omega$. At full-scale V_{in} , resistor R_0 generates a maximum input current of $400 \mu A$, which remains below the $525 \mu A$ coarse currents.

The input buffer circuit converts V_{in} into input current I_0 . When the digital control signal K_0 asserts high potential, input control switch Q_2 closes, directing I_0 to the integrator. The conduction duration corresponds to the signal integration time T selected by the master controller. Either coarse reference current I_1 or I_2 is superimposed on I_0 before entering the integrator. The integrator's output voltage increases or decreases over time (depending on the polarity of the coarse reference current), I_2 (forward coarse current) drives positive integration, while I_1 (reverse coarse current) drives negative integration.

The integrator feeds the integrated signal to the zero-crossing amplifier U_3 . Utilizing the characteristics of diodes, the circuit presents high impedance when the forward current is in the microampere range, while switching to low resistance under larger bias currents. As a result, the amplifier provides high voltage gain for small input signals and unity gain for large signals. This design significantly enhances the amplification factor near the 0-V crossing, thereby improving the sensitivity of subsequent zero-crossing detection.

The signal amplified by the zero amplifier is input to the zero-crossing comparator. When the input voltage is positive, the output saturates at a high logic level; when negative, it drops to a zero level. The comparator output is routed to the master controller for switch circuit control. Q_3 and Q_4 constitute a balancing circuit, driven by complementary control signals. During ADC operation, K_4 is set low, clamping

the junction of R_8 and R_9 to zero potential. When the ADC halts, K_4 is set high, enabling a positive precision current I_3 to flow into the integrator. This current induces a gradual rise in the zero-crossing amplifier's output. Simultaneously, resistor R_8 ($5k\Omega$) generates a balancing current I_B opposing I_3 . At equilibrium, I_3 and I_B become equal in magnitude but opposite in direction, stabilizing the integrator output at a fixed balance voltage V_B (20.5 mV). Consequently, the comparator output settles at this micro-voltage level.

V. INTEGRATED WAVEFORM THEORY AND RESOLUTION LIMIT ANALYSIS

Set the coarse counting frequency to 2 MHz, then the coarse pulse clock cycle T_C is $0.5 \mu s$. With an integrator capacitance of 2.2 nF, the integrator output voltage V_1 can be expressed by equation (1) as follows:

$$V_1 = \frac{I_1 T_C}{C} = \frac{525 \mu A \times 0.5 \mu s}{2.2 nF} = 0.119 V \quad (1)$$

The time t_5 required for coarse current charging to 5V is shown in the following equation (2):

$$t_5 = \frac{V_5 C}{I_1} = \frac{5V \times 2.2 nF}{525 \mu A} = 21 \mu s \quad (2)$$

In the formula: V_5 represents the integrator output of 5V. When reverse charging the capacitor with a fine-tuning current of $4.1 \mu A$ until reaching 0.119 V, the required charging duration t_f is given by the following formula:

$$t_f = 128 \times 0.5 \mu s = 64 \mu s$$

When the input voltage $V_{in} = 0V$, the initial reverse charging current is $525 \mu A$, and the charged voltage V_{17} at a reverse charging duration of $17 \mu s$ is given by:

$$V_{17} = -\frac{525 \mu A \times 17 \mu s}{2.2 nF} = -4.05 V$$

To ensure optimal operation of the ADC integrator across its dynamic range, the circuit must maintain linearity within the $\pm 10 V$ input voltage span. Consequently, the reverse and forward coarse charging durations are configured as $17 \mu s$ and $16 \mu s$, respectively.

When the input voltage is 0V, the initial reverse charging current is $525 \mu A$, and after $17 \mu s$, the reverse charging voltage is $-4.05V$. Similarly, the forward charging current is $-525 \mu A$, and after $16 \mu s$, the forward charging voltage V_{16} is:

$$V_{16} = -\frac{-525 \mu A \times 16 \mu s}{2.2 nF} = 3.81 V$$

When the input voltage is 10 V, the initial reverse charging current I_a is:

$$I_a = I_2 + I_0 = 525 \mu A + 400 \mu A = 925 \mu A$$

Then, after $17 \mu s$, the reverse charging voltage V_a is:

$$V_a = -\frac{925 \mu A \times 17 \mu s}{2.2 nF} = -7.15 V$$

Similarly, when the input voltage is $-10 V$, the initial reverse charging current I_b is:

$$I_b = I_2 + I_0 = 525 \mu A - 400 \mu A = 125 \mu A$$

Then, $17 \mu s$ later, the reverse charging voltage V_b is:

$$V_b = -\frac{125 \mu A \times 17 \mu s}{2.2 nF} = -966 mV$$

In summary, when the integrator is operating at full-scale input, the output voltage swing ranges approximately from $-0.97 V$ to $+7.1 V$ and from $-7.1 V$ to $+0.97 V$. When the input voltage is $0 V$, the voltage swing ranges from $-4.05 V$ to $+3.81 V$ and from $-3.81 V$ to $+4.05 V$. The power supply for the integrator is $\pm 15 V$, ensuring that the integrator always operates within its optimal dynamic range. The bidirectional controllable integration mechanism proposed in this paper extends the positive and negative bipolar input range to $-10 V$ to $+10 V$, overcoming the limitation of traditional unidirectional integration without relying on external signal conditioning circuits. The theoretical waveform of the full-scale input integration described above is shown in Figure 3.

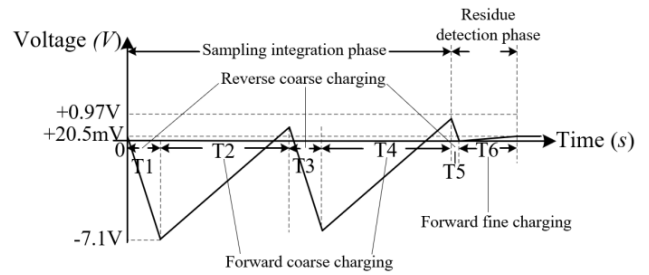


FIGURE 3. Theoretical integration waveform diagram under full-range input.

In the theoretical waveform diagram of the 10 V full-scale input integration, intervals T_1 and T_3 represent the reverse coarse charging time of approximately $17 \mu s$. Upon reaching $-7.1 V$, the system switches to a $16 \mu s$ forward coarse charging phase. If no zero-crossing occurs within this period, it continues to charge with forward coarse current for another $16 \mu s$ until a zero-crossing occurs during one of the $16 \mu s$ periods of forward coarse current charging, as depicted by intervals T_2 and T_4 , with the maximum voltage of the forward coarse charging being below $+0.97V$. After completing the forward coarse charging, start the reverse coarse charging and repeat the above process. At the conclusion of the signal integration phase (e.g., interval T_5), if the integrator output is positive when the measured signal ends, discharge using reverse coarse current until the integrator output crosses zero, followed by a slow charge with a forward fine current until the integrator output reaches zero; Conversely, if the integrator output is negative when the measured signal ends, a forward coarse current charges it to positivity, then a reverse coarse current discharges it to zero, and finally a forward fine current slowly charges it to positive zero-crossing. At this

point, Maintaining the forward fine current and activating the balance switch stabilizes the integrator output at 20.5 mV, concluding one complete analog-to-digital conversion cycle.

As mentioned above, the switching states of Q_1 and Q_2 are governed by control signal K_0 . When K_0 is introduced at a high potential, the system selects the integration time T for the signal, sets the coarse forward/reverse reference current to $525 \mu\text{A}$, and sets the input full-scale $\pm 10 \text{ V}$ signal charge/discharge current I_0 to $400 \mu\text{A}$. Assuming that T is the signal integration time (NPLC) of input current I_0 , T_x is the coarse counting time, then T_x can be obtained from formula (3):

$$T_x = \frac{I_0}{I_1} T = \frac{400 \mu\text{A}}{525 \mu\text{A}} T = 0.76 T \quad (3)$$

At a coarse counting frequency of 2 MHz, the coarse counting time (corresponding to the coarse reference access time) yields coarse counting values of 304 (8 bits), 3040 (11 bits), 30400 (14 bits), and 304000 (18 bits). Given that the fine counting discharge speed is approximately 128 times (7 bits) slower than the coarse counting speed, and the fine counting frequency is increased to 16 MHz (3-bit resolution gain), the overall resolution improvement of fine counting amounts to 10 bits.

In the hierarchical design of “high-order sampling integration and low-order residual detection” adopted in this paper, it is assumed that the high-order quantization series is N_{high} , the low-order quantization series is N_{low} , the coarse quantization counting frequency is f_{high} , and the fine quantization counting frequency is f_{low} , the high-order integration unit time is t_{high} , and the low-order integration unit time is t_{low} .

Since the fine counting frequency f_{low} during low-order detection is 8 times the coarse counting frequency f_{high} , and the coarse charging current is 128 times the fine charging current. Therefore, the time required for the low-order integration time t_{low} to reach the same potential is 128 times that of the high-order integration time t_{high} . Thus, the low-order quantization sequence N_{low} is shown in equation (4) below:

$$N_{\text{low}} = f_{\text{low}} t_{\text{low}} = 8 f_{\text{high}} \times 128 t_{\text{high}} = 1024 N_{\text{high}} \quad (4)$$

From the above equation (4), it can be observed that the minimum quantization step of the high order is 1024 times that of the low order, and the corresponding improvement in the effective number of bits (ENOB) can be expressed by formula (5) as follows:

$$\Delta \text{ENOB} = \log_2(1024) = 10 \text{ bit} \quad (5)$$

From equation (5) above, it is evident that the introduction of low-order residual detection (essentially extending the integration time and enhancing the counting density) can improve the effective resolution by 10 bits.

In summary, the resolutions corresponding to the signal integration times (NPLC) are 18 bits, 21 bits, 24 bits,

and 28 bits, respectively, and the decimal resolutions are 5.5 digits, 6.5 digits, 7.5 digits, and 8.5 digits, respectively.

Any measurement of the input signal V_{in} involves three conversions: the measured voltage V_x , the ground signal V_g , and the highly stable precision reference voltage V_r . The measurement of ground-referenced voltage V_g serves to reject common-mode ground voltage in signal V_x , whereas the V_r reference measurement enables per-sample calibration against voltage standards to compensate for temperature-induced parametric drift in ADC components.

During normal operation of the ADC, the following count values are involved: the coarse count N corresponding to the reverse coarse current I_1 , the coarse count M corresponding to the forward coarse current I_2 , the count Z corresponding to the total integration time T_Z of the input signal under varying signal integration durations, and the fine count J corresponding to the forward fine current I_3 .

The ADC operation starts and ends at the balancing voltage V_B (20.5 mV). The difference between the forward integral charge and the reverse integral charge is only the 20.5 mV voltage stored by the initial 2.2nF capacitance, which is the equilibrium charge Q_B . Let T_N denote the reverse coarse time, T_M the forward coarse time, T_Z the input integration time, and T_J the fine forward time. From the charge balance equation, the measured voltage V_x can be obtained from the following equation (6):

$$I_1 T_N + I_x T_z = I_1 T_M + I_3 T_J + Q_B \quad (6)$$

Let the forward and reverse coarse counting clock cycles be t_M and t_N respectively, with $t_M = t_N$. Define t_J is the fine counting clock cycle. As mentioned earlier, t_M should be 8 times t_J , and I_1 is approximately 128 times I_3 . Therefore, we can obtain:

$$I_1 t_N N + \frac{V_x}{R_x} t_N Z = I_1 t_M M + I_3 t_J J + Q_B$$

Transposing terms yields formula (7):

$$\frac{V_x t_N Z}{R_x} = (M_x - N_x) I_1 t_N + I_3 t_J J_x + Q_B \quad (7)$$

When the input voltage is GND, The following relationship (8) can be derived:

$$\frac{V_g t_N Z}{R_x} = (M_g - N_g) I_1 t_N + I_3 t_J J_g + Q_B \quad (8)$$

When the input voltage equals the reference voltage, The following relationship (9) can be derived:

$$\frac{V_r t_N Z}{R_x} = (M_r - N_r) I_1 t_N + I_3 t_J J_r + Q_B \quad (9)$$

Formula (10) is derived from equations (7) to (8) as follows

$$\frac{(V_x - V_g) Z}{R_x} = (M_x - N_x - M_g + N_g) I_1 + \frac{(J_x - J_g) I_3}{8 \times 128} \quad (10)$$

Formula (11) is derived from equations (9) to (8) as follows

$$\frac{(V_r - V_g)Z}{R_x} = (M_r - N_r - M_g + N_g)I_1 + \frac{(J_r - J_g)I_3}{8 \times 128} \quad (11)$$

where $V_x - V_g$ is the actual measured voltage value V_{XT} , and $V_r - V_g$ is the precision reference voltage. From equations (10)/(11), we can obtain:

$$V_{XT} = \frac{V_r [2^{10} (M_x - N_x - M_g + N_g) + (J_x - J_g)]}{2^{10} (M_r - N_r - M_g + N_g) + (J_r - J_g)} \quad (12)$$

Equation (12) shows that the fine count resolution is 1024-fold higher than the coarse count, corresponding to a 10-bit enhancement. The final measured voltage V_{XT} is only related to the positive and negative coarse count values, the positive fine count value, and the precision reference voltage.

VI. INTEGRAL CONTROL PROCESS DESIGN

The system monitors the voltage across the integrating capacitor to enable real-time control of reference level switching. Among them, the ADC control is divided into the sampling integration phase and the residual detection phase, with integral zeroing performed synchronously at the end of the residual detection phase [29].

The sampling and integration flowchart of the system is shown in Figure 4. Upon receiving the start signal for sampling and integration, the system first performs polarity determination on the input signal. When the input signal is positive, it executes a 17 μ s timed inverting integration and a 16 μ s timed forward integration. Thereafter, it resumes timed inverting integration until the integrator crosses zero during an integration cycle. If the input signal remains unterminated after this integration cycle, it performs timed forward integration again and repeats the process. When the input signal is negative, it conducts one timed inverting integration followed by timed forward integration until a zero-crossing event occurs in the integrator during an integration cycle. If the input signal is still not terminated after this integration, it initiates timed forward integration again and repeats the aforesaid process. The input signal is disconnected when its integration time T elapses, marking the end of sampling and integration.

Once the sampling integration is completed, the system immediately enters the residual detection phase. The residual detection flowchart of the system is shown in Figure 5. The polarity of the residual voltage is determined by detecting the state of the zero crossing comparator at the moment of starting the residual detection. When the residual voltage is positive, carry out reverse coarse charging until the integrator has zero crossing, start positive fine charging at the moment of zero crossing signal, and open the balance switch to adjust the integrator to zero synchronously. When the residual voltage is negative, the integrator is first charged to a positive voltage using a forward coarse current, followed by reverse coarse charging until the integrator crosses zero. Similarly,

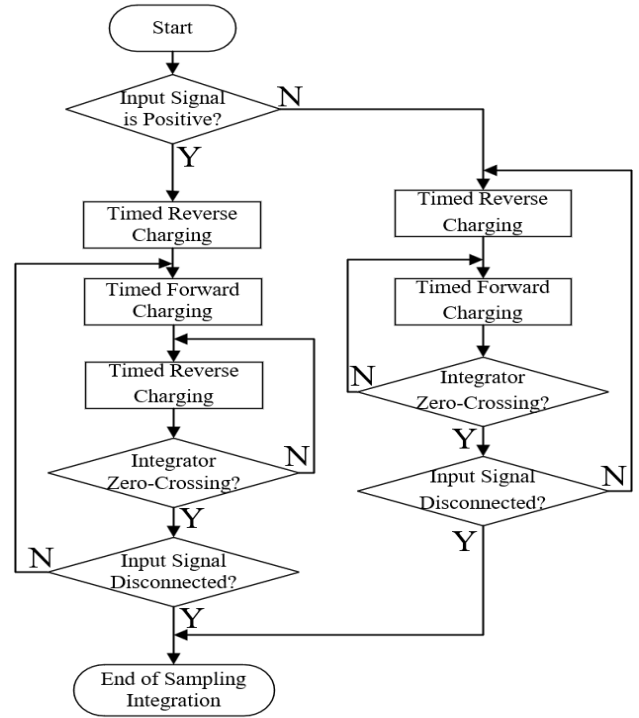


FIGURE 4. Flowchart of the sampling-integration phase.

at the moment the zero-crossing signal appears, forward fine charging is activated, and the balance switch is opened to zero the integrator, preparing it for the next conversion.

As can be seen in Figures 4 and 5 above, the overall workflow of the sampling-integration phase and residual detection phase is illustrated. Next, the timing relationships of key signals in different phases will be elaborated in detail with reference to the control Signal timing diagram in Figure 6 below.

As can be seen in Figure 6 above, it is not difficult to see that before the input signal control switch is turned on, the integrator output is maintained at the balance potential by the forward fine current and the balance circuit. Once the input signal control switch is turned on, it indicates the start of sampling integration. First, the reverse coarse charging switch is turned on for a reverse coarse charging period of approximately 17 μ s, followed by several forward coarse charging cycles until the comparator crosses zero. As shown in Figure 6 example, the zero crossing occurs in the third forward coarse charging cycle. At this point, since the input signal control switch has not yet been turned off, the above-mentioned forward/reverse coarse charging cycles are repeated. When the input signal control switch is turned off, it means that the sampling integration is over and the residual detection starts. In the above timing diagram example, since the comparator is at a high level, the voltage involved in the analysis is positive, so the reverse coarse charging begins until the zero crossing. Subsequently, the forward fine charging switch and the balancing circuit switch are activated and remain active until the integrator output reaches the balanced potential.

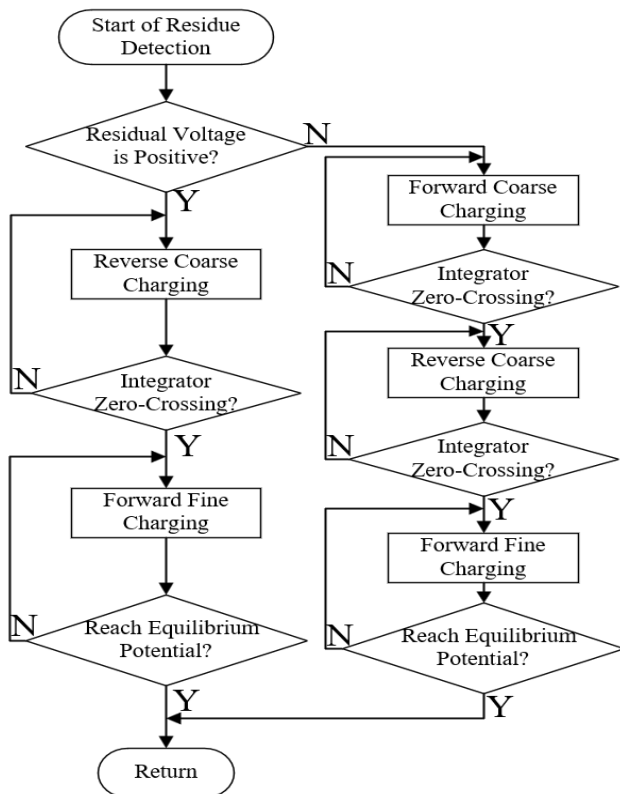


FIGURE 5. Flowchart of residual detection phase.

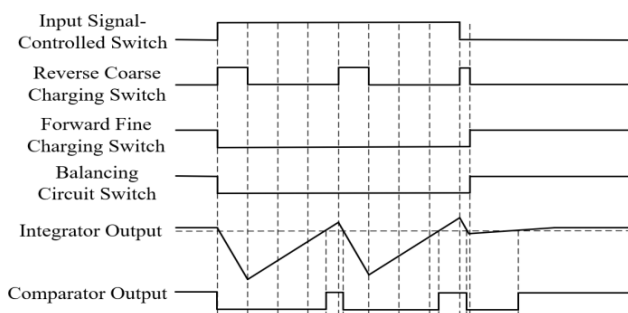


FIGURE 6. Control signal timing diagram.

VII. EXPERIMENTAL RESULTS AND ANALYSIS

Experimental voltage waveforms of the proposed bidirectional multi-slope integrating ADC under zero-input condition and full-scale excitation are captured in Figure 7 and Figure 8, respectively. The four vertically aligned traces in these oscillograms, sequentially labeled from top to bottom as Signal 1 through Signal 4, represent critical nodal responses during the conversion sequences.

In Figures 7 and 8, Waveform 1 corresponds to the input switch control signal, while Waveform 2 represents the reverse integration control signal. Signal 3 exhibits the output voltage variation of the integrator, and Signal 4 depicts the output voltage variation of the zero-crossing comparator. The voltage reference for Signals 1 and 2 is set to 10 V/div, the voltage reference for Signals 3 and 4 is set

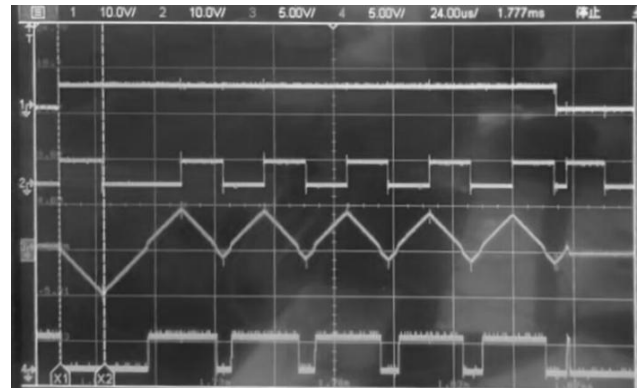


FIGURE 7. Dynamic response to 0-V input.

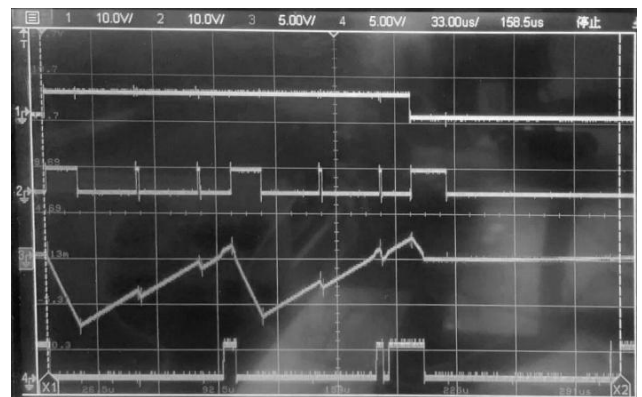


FIGURE 8. Dynamic response to full-scale input.

to 5 V/div, the time base in Figure 7 is set to 24 μ s/div, and the time base in Figure 8 is set to 33 μ s/div.

Under both zero-input and full-scale excitation conditions, the integrator is precisely regulated by the input switching signal and reverse integration control signal, with experimental results exhibiting excellent agreement with theoretical integration waveforms. The integrator output demonstrates negligible overshoot or ringing at slope commutation points, maintaining voltage stability without observable distortion, thereby validating effective suppression of nonlinear errors. The zero-crossing comparator output triggers transitions precisely when the integrator crosses the zero level, with steep rising and falling edges indicating superior dynamic response capability.

To more intuitively demonstrate the performance advantages of the proposed design, Table 2 compares its key parameters with those of existing similar ADC architectures, including effective number of bits (ENOB), power line frequency rejection ratio (PLFRR), and integral nonlinearity (INL), so as to highlight the advantages of the proposed design.

As shown in Table 2 above, the proposed ADC in this work implements 24 bit ENOB, which is comparable to the advanced TI-ADS1262. However, unlike the TI-ADS1262 that relies on high oversampling rates and complex digital

TABLE 2. Key parameters comparison of various ADC types.

ADC Type	Effective Number of Bits (ENOB)	Power Line Frequency Rejection Ratio (PLFRR)	Integral Nonlinearity (INL)
This Work	24-bit	110dB@50HZ	± 1 ppm
[5] (SAR ADC)	21-bit	80dB@50HZ	± 2 ppm
[6] (Δ - Σ ADC)	22-bit	110dB@50HZ	± 4 ppm
TI-ADS1262 (Δ - Σ ADC)	24-bit	110dB@50HZ	± 3 ppm

TABLE 3. Full-scale voltage test data.

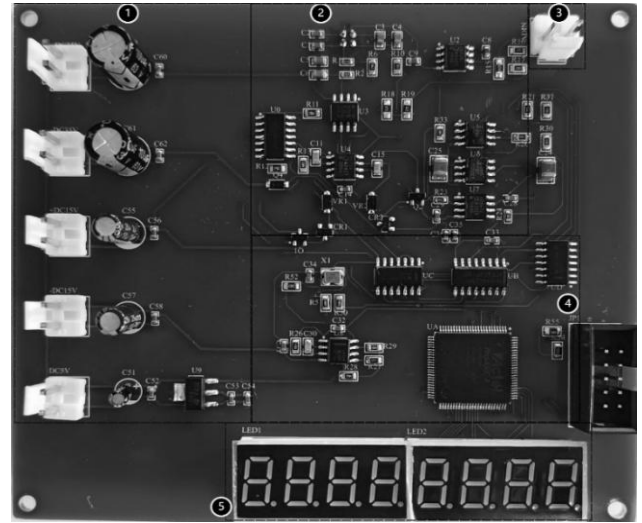
Measurement Sequence	Measured Value / V	Measurement Sequence	Measured Value / V
1	10.000 006	6	10.000 007
2	10.000 006	7	10.000 007
3	10.000 008	8	10.000 005
4	10.000 006	9	10.000 006
5	10.000 007	10	10.000 006

filtering for noise suppression, the “segmented time quantization mechanism” introduced in this design subdivides the analog voltage into high-precision time interval measurements, thus avoiding the constraints imposed by operational amplifier noise and switching charges on high resolution in the Σ - Δ architecture. In addition, unlike the Δ - Σ ADC in [6], which depends on an additional digital filter to achieve 110dB PLFRR, this design realizes a PLFRR of 110dB@50Hz at the hardware level by locking the integration period to an integer multiple of the power frequency period. Furthermore, the integration-stage error cancellation mechanism adopted in this work naturally offsets fixed errors such as integration capacitor leakage and operational amplifier offset voltage, breaking through the 2ppm linearity bottleneck of the SAR ADC in [5]. The results in Table 2 confirm the advantages of the proposed ADC in terms of resolution, anti-interference capability, and nonlinear error suppression.

Figure 9 depicts the prototype of the bidirectional multi-slope integrating ADC. After 30-min instrument warm-up (Fluke 5730A calibrator), the input voltage port of ADC was connected to its precision DC source. With 10-V input excitation and fixed 1 PLC integration time, ten successive conversions yielded the data in Table 3, where maximum deviation < 9 LSB confirms excellent short-term stability.

VIII. CONCLUSION

This paper presents a bidirectional multi-slope integrating ADC, detailing its hardware design, resolution computation, and software design. By adaptively configuring signal integration periods, this converter enables dynamic resolution selection to achieve flexible adaptation across diverse



①- Power Supply Module ②- Sampling Integration Module
③- Input Interface ④- Control Module ⑤- Display Module

FIGURE 9. Physical implementation of bidirectional multi-slope integral ADC.

applications. Theoretical analysis confirms it attains an 8.5-digit resolution at a 10 PLC integration setting.

It should be noted that the multi-slope integrating ADC designed in this paper has certain constraints on the architecture characteristics: it relies on precise regulation of the inverse coarse reference current, forward coarse reference current, and forward fine reference current to integrate with the input signal. Consequently, stringent requirements are imposed on the stability and symmetry of the reference power supplies generating these currents—even minor mismatches may introduce additional nonlinear errors. Additionally, while the dynamic resolution design enables flexible adaptation to diverse applications, an intrinsic trade-off exists between conversion speed and accuracy. To achieve high-resolution conversion, the integration time and counting density must be increased, thereby extending the overall conversion time. Conversely, in high-speed mode, the resolution is inevitably compromised.

Experimental results demonstrate that with a 1 PLC signal integration time, the 10 V full-scale DC voltage test meets the 7.5-digit resolution requirement and exhibits excellent short-term stability. The ADC circuit offers significant practical utility in low-speed, high-resolution systems, particularly in medical diagnostic devices and precision calibration systems requiring metrological-grade accuracy.

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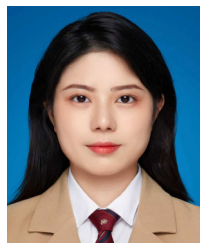
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